PROJECT POSTER

CRICKET PLAYER PERFORMANCE PREDICTION

using Ridge and Lasso Regression

PRESENTED BY

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GUIDE

WHY IT MATTERS

Selectors, IPL franchises and fantasy players need a reliable estimate of a cricketer's value. We turn match statistics into a single 0 to 100 performance rating, using regularised regression that copes with overlapping stats such as strike rate and boundary percentage.

DATASET AND FEATURES

Batting
Average, strike rate, last 5 runs, boundary %, balls faced, 50s / 100s

Bowling
Economy, bowling average, last 5 wickets, dot-ball %

Context
Age, matches, fitness, catches, format, pitch, opposition

RIDGE (L2)

$$\text{Loss} + \alpha \Sigma \beta^2$$

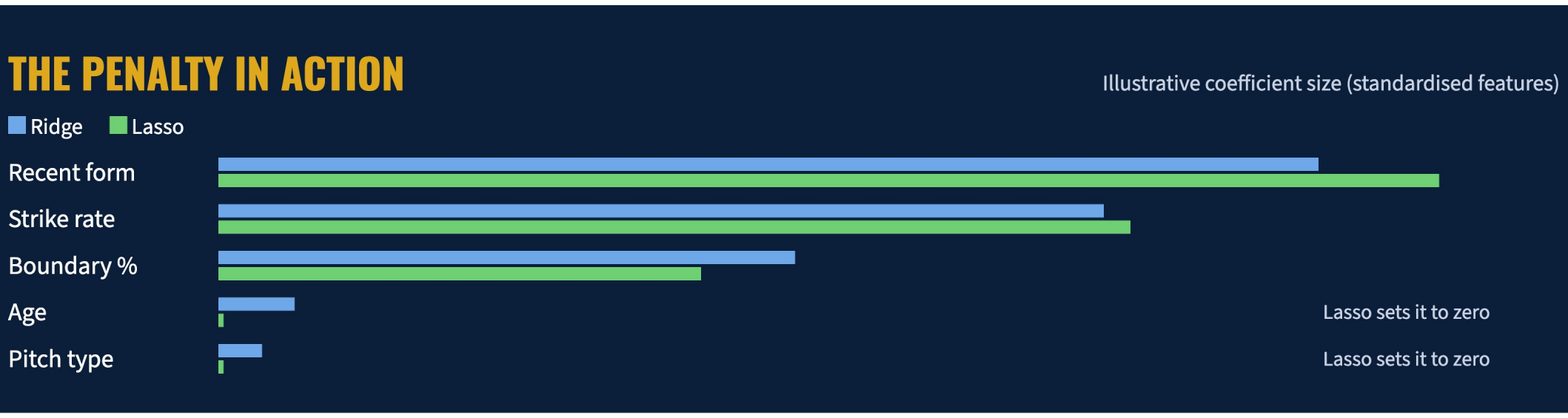
Shrinks every coefficient smoothly and keeps all features. Stays stable when features are correlated.

LASSO (L1)

$$\text{Loss} + \alpha \Sigma |\beta|$$

Pushes weak coefficients to exactly zero, giving automatic feature selection and a compact, explainable model.

THE PENALTY IN ACTION



METHODOLOGY

1 Match data → 2 Cleaning → 3 Feature engineering → 4 Scaling → 5 Train / test split → 6 RidgeCV / LassoCV → 7 Evaluation → 8 Prediction

EVALUATION

MAE average rating error
RMSE penalises large misses
R^2 variation explained
Linear, RidgeCV and LassoCV are scored on the same held-out test set.

KEY INSIGHTS

Recent form and strike rate carry the most weight. Ridge stays stable with correlated stats. Lasso drops noisy features. Alpha is chosen by cross-validation.

CONCLUSION

Regularised regression predicts cricket performance reliably. Future work: real ball-by-ball and IPL data, Elastic Net, and a live fantasy-team predictor.

Guide: **Dr. M. Rani**
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